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Construction of a skeletal oxidation mechanism of *n*-pentanol by integrating decoupling methodology, genetic algorithm, and uncertainty quantification

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ABSTRACT

Pentanol has attracted increasing attentions in recent years due to its ability to reduce the pollution emissions of engines and the dependence on fossil fuels. A skeletal oxidation mechanism composed of 47 species and 177 reactions is first developed for *n*-pentanol based on the decoupling methodology in this study. Then, the rate constants of the reactions in the fuel-related sub-mechanism are automatically optimized by using the genetic algorithm to reproduce the ignition delay times in shock tubes and rapid compression machines, and the major species concentrations in jet-stirred reactors. The final mechanism is determined based on the method of uncertainty minimization using polynomial chaos expansions by comparing the predicted uncertainty of the optimized mechanisms with available experimental data in shock tubes, rapid compression machines, and jet-stirred reactors. The final *n*-pentanol mechanism is validated against measurements in shock tubes, rapid compression machines, jet-stirred reactors, and premixed laminar flames over low-to-high temperatures. Good agreements between the measured and predicted results are obtained for various reactors. Due to the compact size and the reliable performance, the final mechanism is capable of well reproducing the combustion and emission behaviors of *n*-pentanol in a homogeneous charge compression ignition engine in coupling with a three-dimensional Computational Fluid Dynamics (CFD) model.

1. Introduction

As the primary propulsion for ground transportation, internal combustion engines require higher fuel economy and lower pollution emissions to meet the energy crisis and the stringent emission regulations. Biofuels, such as bio-alcohols, have attracted increasing attentions as additive or pure fuel in internal combustion engines because of the capacity of improving energy security and reducing pollutant emissions [1–3]. Among various bio-alcohols, *n*-pentanol is regarded as one of the most potential next-generation biofuels due to its similar density and viscosity to diesel and competitive chemical properties compared with the other alcohols [4,5]. Li et al. [6,7] found that the NO_x and soot emissions can be reduced simultaneously for the engine fueled with pure pentanol or pentanol/diesel blends at low and middle loads compared to diesel, while the increased NO_x emissions were found at high loads. To explain the influence of pentanol on the engine performance and emissions, an in-depth understanding of the oxidation and combustion behaviors of pentanol is crucial.

By adding the *n*-pentanol sub-mechanism into the C_1–C_4 alcohol mechanisms [8–11], Togbe et al. [12] first proposed a detailed *n*-pentanol oxidation mechanism including 261 species and 2099 reactions. The mechanism was employed to predict the species concentrations in jet-stirred reactors (JSRs) and laminar flame speed covering the temperature (T) range of 770–1220 K, pressure (p) range of 1–10 atm, and equivalence ratio (φ) range of 0.35–4.0, and the predicted results showed good agreement with experimental data. Tang et al. [13] refined the rate constants of the *n*-pentanol sub-mechanism in the Togbe et al. mechanism [12] to improve the predictions on the ignition delay times at high temperatures. Li et al. [14] also modified the Togbe et al. mechanism [12] based on the sensitivity analysis of the laminar flame speed. The results indicated that the modified mechanism well reproduced the flame propagation characteristics of *n*-pentanol.

Kohler et al. [15] used the Reaction Mechanism Generator (RMG) to automatically generate a detailed oxidation mechanism for *n*-pentanol. The mechanism satisfactorily reproduced the ignition delay time, laminar flame speed, and species concentration in premixed laminar flame at high temperatures over $p = 0.02$–2.6

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atm and $\varphi = 0.25$–1.8. Recently, Nativel et al. [16] proposed a high-temperature kinetic mechanism to describe the ignition delays and flame propagation characteristics of n-pentanol at $T = 1100$–1500 K, $p = 1$–7.5 atm, and $\varphi = 0.25$–1.8, and the mechanism included 230 species and 7885 reactions.

Using the same methodology and framework as that for the mechanism of butanol isomers [17], Heufer et al. [18] built a comprehensive n-pentanol oxidation mechanism with 599 species and 3010 reactions. The mechanism was validated against the experimental data in shock tubes, rapid compression machines (RCMs), JSRs, and premixed laminar flames over ranges of $T = 640$–1200 K, $p = 9$–30 atm, and $\varphi = 0.7$–1.6. Furthermore, Wang et al. [19] proposed a detailed mechanism composed of 314 species and 1602 reactions to describe the pyrolysis and oxidation of n-pentanol covering low to high temperatures. The predictions of the mechanism on ignition delay time, species concentration, and laminar flame speed were in good agreement with the measured data.

Although several n-pentanol oxidation mechanisms have been proposed in recent years, the detailed mechanisms usually consist of hundreds of species and thousands of reactions to describe all the relevant pathways. It needs long computational time to run a code which integrates the detailed mechanism into a three-dimensional computational fluid dynamics (3D CFD) model. Thus, reduced/skeletal n-pentanol mechanisms are required.

In recent years, various mechanism reduction techniques have been proposed. Based on the reduction principle, the techniques can be categorized into direct reduction method, lumping method, and time-scale analysis [20]. In the direct reduction method, unimportant species and reactions are eliminated from the detailed mechanism. The typical direct reduction methods include the directed relation graph (DRG) and its derived methods [21–23], sensitivity analysis [24,25], principal component analysis [26,27], etc. In contrast, species with the same functional group are lumped into one representative species in the lumping method [28,29]. The time-scale analysis is based on the quasi-steady-state and partial equilibrium hypothesis [30], in which the time evolutions of the fast species and reactions are described by algebraic equations. The representative methods of the time-scale analysis include intrinsic low-dimensional manifolds (ILDM) [31] and computational singular perturbation (CSP) [32,33].

It has been found that the accuracy of the reduced mechanisms obtained through the above methods strongly rely on the reduction objective [25,34,35]. Moreover, the size of the reduced mechanisms is still large for 3D CFD model [36]. To further reduce the size and improve the performance of the reduced mechanisms, a decoupling methodology was proposed [37,38], in which a chemical mechanism is composed of a skeletal fuel-specific C_4–C_n sub-mechanism, a reduced C_2–C_3 sub-mechanism, and a detailed $H_2/CO/C_1$ sub-mechanism. An amount of validated results demonstrate that the skeletal mechanisms developed by the decoupling methodology are capable of satisfactorily reproducing the ignition delay times, species evolutions, and laminar flame speeds of pure component, mixture, and surrogate fuel under a wide operating range, and the skeletal mechanism can also be applied in coupling with 3D CFD models to predict the ignition and emission characteristics of practical engines due to their compact sizes [37,39].

An important procedure in the mechanism development using the decoupling methodology is the optimization of the reaction rate constants in the C_4–C_n sub-mechanism [37]. In a previous work [37], the optimization process was performed by manual iteration based on the sensitivity analysis and the path analysis, which is time-consuming and strongly relies on the experience of the handlers.

In the present work, a new reduced n-pentanol mechanism is proposed by coupling the decoupling methodology, genetic algorithm, and uncertainty quantification. First, a skeletal n-pentanol mechanism is constructed based on the decoupling methodology, in which the C_2–C_3 sub-mechanism is updated to improve the predictions on the C_2H_2 and C_2H_4 species concentrations. Second, the rate constants in the fuel sub-mechanism are optimized with the Non-dominated Sorting-based Genetic Algorithm II (NSGA-II) to reproduce the ignition delay times in shock tubes and RCMs, and the major species concentrations in JSRs over wide operating conditions, consequently, a set of optimized mechanisms are ob-

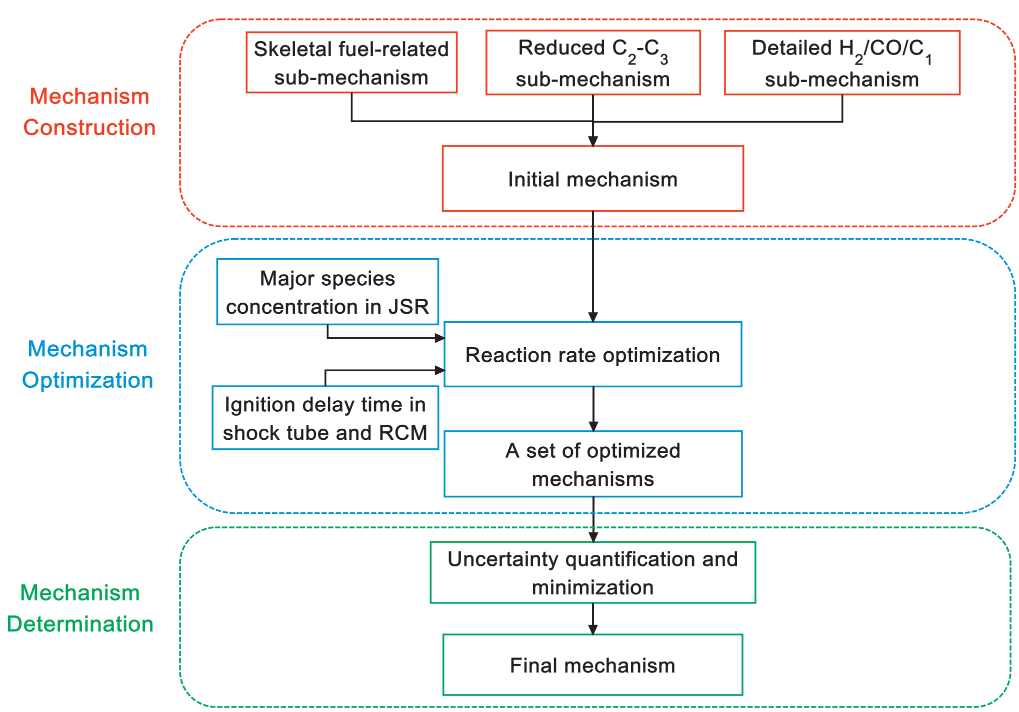

Fig. 1. Schematic diagram of mechanism construction, optimization, and determination.

tained by NSGA-II. The final mechanism is determined by assessing the optimized mechanisms based on the Method of Uncertainty Minimization using Polynomial Chaos Expansions (MUM-PCE). Third, the final mechanism is validated against the experimental data in shock tubes, RCMs, JSRs and premixed laminar flames. Due to its compact size, the final mechanism is also used to predict the combustion and emission characteristics of a practical engine fueled with *n*-pentanol. Finally, major conclusions are presented.

2. Mechanism development

In this section, the processes of the construction of the skeletal mechanism, the optimization of the reaction rate, and the determination of the final mechanism are presented, as displayed in Fig. 1. First, the initial mechanism is constructed using the decoupling methodology. Then, the rate constants of the reactions in the skeletal fuel-related sub-mechanism are optimized to reproduce the major species concentrations in JSRs and the ignition delay times in shock tubes and RCMs with the genetic algorithm, and a set of optimized mechanisms can be obtained. Finally, the final mechanism is determined based on the uncertainty analysis.

2.1. Mechanism construction

Due to the effectiveness and reliability, the decoupling methodology is employed to construct the *n*-pentanol skeletal mechanism in this study. The decoupling methodology has been described in detail in Ref. [37], and only a brief introduction is presented herein. In the decoupling methodology, a skeletal mechanism is constructed including three parts, i.e., a skeletal fuel-related C_4–C_n sub-mechanism, a reduced C_2–C_3 sub-mechanism, and a detailed $H_2/CO/C_1$ sub-mechanism. The skeletal fuel-related sub-mechanism, which is composed of the high-temperature, low-temperature, and functional group reactions, is introduced to predict the ignition delay time and fuel consumption. Due to the inclusion of the functional group reactions, the skeletal mechanism is capable of capturing the unique oxidation characteristics of different fuels, such as the early formation of carbon dioxide (CO_2) for methyl decanoate [40] and the weak negative temperature coefficient (NTC) behavior of *n*-butanol [41].

Meanwhile, the detailed $H_2/CO/C_1$ sub-mechanism is employed to reproduce the flame propagation characteristics and the evolution of the major small species (e.g., H_2, H_2O, CO, CO_2, etc.). The reduced C_2–C_3 sub-mechanism is utilized to associate the skeletal fuel-related sub-mechanism with the detailed $H_2/CO/C_1$ sub-mechanism. Thereby, the skeletal mechanism developed by the decoupling methodology is capable of well predicting the ignition delay time, major species evolution, and flame propagation of both large-hydrocarbon and oxygenated fuels. Moreover, the compact size of the skeletal mechanism makes it possible to be integrated with 3D CFD simulation.

2.1.1. Skeletal fuel-related C_4–C_n sub-mechanism

$nC_5H_{11}OH + X = nC_5H_{10}OH + HX$ (R1–R4)

X represents H, O_2, OH and HO_2

Based on the path analysis of the detailed *n*-pentanol mechanisms [12, 16, 18], the major reaction paths of the *n*-pentanol oxidation mechanism developed using the decoupling methodology are illustrated in Fig. 2. The fuel molecule is first consumed to form $nC_5H_{10}OH$ through the H-atom abstraction reactions by H, OH, O_2, and HO_2 in reactions R1–R4 [12, 15, 18]. At low temperatures, there are two major reaction paths. One is $nC_5H_{10}OH \rightarrow C_5H_{10}OHO_2 \rightarrow C_5H_9OHOOH \rightarrow C_5H_9OHOOHO_2 \rightarrow C_5OHket \rightarrow$ low-temperature branch, which is similar to the low-temperature reactions of *n*-alkanes with $R \rightarrow RO_2 \rightarrow QOOH \rightarrow O_2QOOH \rightarrow$ ketohdroperoxide $\rightarrow$ low temperature branch. Due to the influence of the alcohol group, $nC_5H_{10}OH$ also reacts with O_2 to produce nC_4H_9CHO (R10). The reaction competes with reaction R5, leading to the lower reactivity of *n*-pentanol than that of *n*-pentane at low temperatures [18, 42]. nC_4H_9CHO consequently reacts with OH and HO_2 to produce nC_4H_9CO, which then decomposes to C_0–C_3 species by reaction R13 [12]. At high temperatures, $nC_5H_{10}OH$ undergoes the β-scission reaction to form CH_3CHO (R14) [12]. CH_3CHO then transforms into $CO + CH_4$ and $HCO + CH_3$ via the decomposition reactions R17 and R18. Another main consumption path of CH_3CHO is the H-atom abstraction reactions (R15 and R16) to yield CH_3CO [12, 43], which subsequently decomposes to CH_3 and CO through reaction R19.

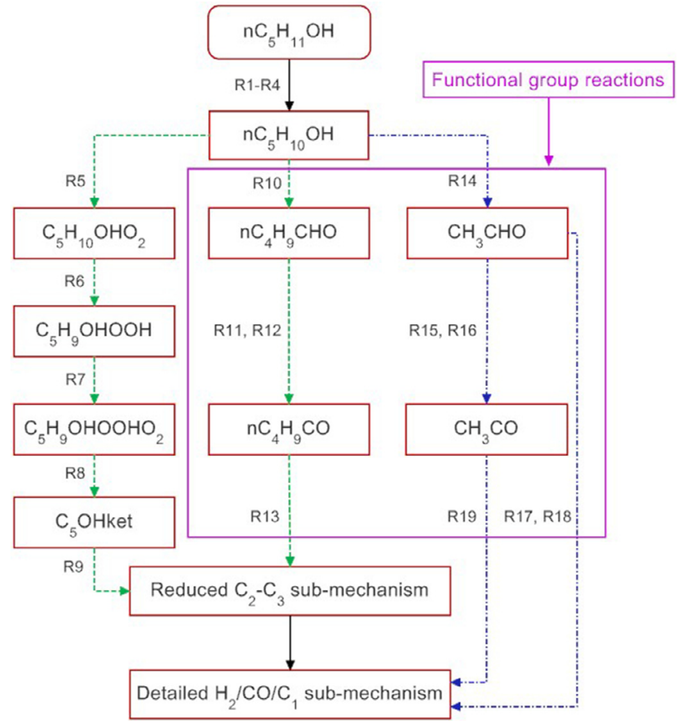

Fig. 2. Major reaction paths of *n*-pentanol mechanism over low-to-high temperatures (green and dashed lines: low-temperature reaction path; blue and dashdotted lines: high-temperature reaction path). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article).

Because of the special molecular structure of *n*-pentanol, a large amount of isomers for C_4–C_5 species exist during the oxidation process of *n*-pentanol. To reduce the size of the skeletal mechanism, only one representative species with the lowest bond dissociation energy in each species group is retained in the C_4–C_n sub-mechanism. For example, α-$C_5H_{10}OH$ is used to represent the $nC_5H_{10}OH$ radicals involving the five isomers because of its lowest bond dissociation energy among the five isomers [18].

2.1.2. Reduced C_2–C_3 sub-mechanism

In our previous work [37], the reduced C_2–C_3 sub-mechanism used in the decoupling methodology is taken from Patel et al. [44], which was developed focusing only on the ignition characteristics of fuels, leading to its poor performance on predicting the C_2 and C_3 species concentrations [45]. It is found that the C_2 and C_3 species, such as C_2H_2 and C_2H_4, play a major role in the polycyclic aromatic hydrocarbons (PAH) and soot formation [46]. To

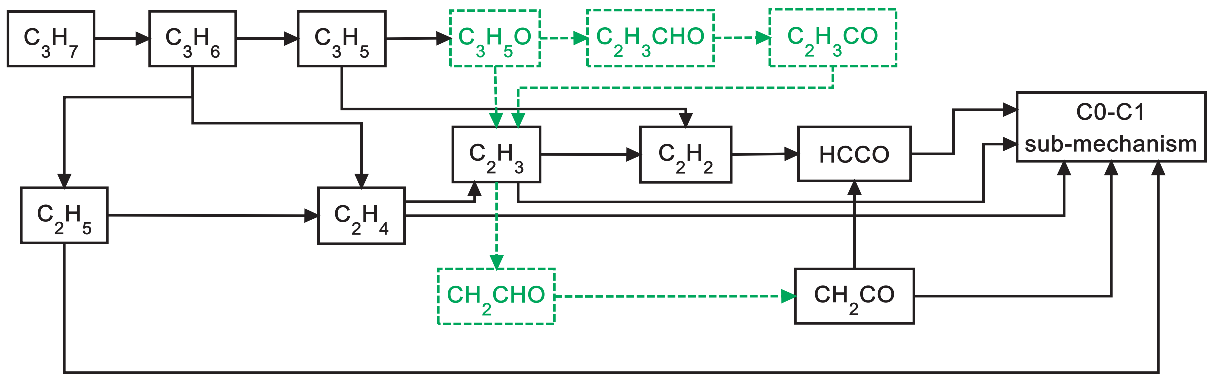

Fig. 3. Reaction paths of C_2–C_3 sub-mechanism (green and dashed lines are the new ones introduced in this study). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article).

improve the accuracy of the final mechanism, an updated C_2–C_3 sub-mechanism is constructed based on the path analysis on C_2H_2 and C_2H_4 species in JSRs, as well as on the sensitivity analysis of laminar flame speed and ignition delay time of the detailed *n*-pentanol oxidation mechanisms [12, 18] under the operating conditions of low-to-high temperatures and high pressures. The reaction paths of the updated C_2–C_3 sub-mechanism are shown in Fig. 3.

The path analysis on C_2H_2 in JSRs indicates that C_3H_5O is an important intermediate affecting the formation of C_2H_2. Thus, the reactions involving the production and consumption of C_3H_5O are added into the C_2–C_3 sub-mechanism. The C_3H_5O radical is formed via the reaction $C_3H_5 + HO_2 = C_3H_5O + OH$ (R26). Then, C_3H_5O is consumed through pyrolysis reactions to form $C_2H_3CHO + H$ (R29) and $C_2H_3 + CH_2O$ (R30). Subsequently, C_2H_3CHO undergoes the H-atom abstraction reactions (R31 – R33) to form C_2H_3CO, which then decomposes into C_2H_3 and CO (R34). For the C_2H_3 radical, the reaction system of $C_2H_3 + O_2$ exhibits high sensitivity coefficients to ignition delay time and laminar flame speed [14, 16], in which four reactions are included, i.e., $C_2H_3 + O_2 = CH_2O + HCO$, $C_2H_3 + O_2 = CH_2CHO + O$, $C_2H_3 + O_2 = C_2H_3O_2$, and $C_2H_3 + O_2 = CH_2O + CO + H$. Mebel et al. [47] indicated that the reaction $C_2H_3 + O_2 = C_2H_3O_2$ was only important at room temperature. Thus, it is not included in the present mechanism.

To reduce the size of the final mechanism, many reaction paths are removed from the $C_2 - C_3$ sub-mechanism. Moreover, the $C_0 - C_4$ mechanism is updated continuously in recent years with the increasing improvement on the chemical kinetic theory [48–51]. Thus, the $C_0 - C_3$ sub-mechanism in the decoupling methodology will be further updated in future work.

The detailed H_2/CO/C_1 sub-mechanism used in this study is taken from the work of Burke et al. [52, 53] because of its reliable performance. By combining the C_4–C_5, C_2–C_3, and H_2/CO/C_1 sub-mechanisms, the final *n*-pentanol skeletal mechanism is constructed including 47 species and 177 reactions.

2.2. Mechanism optimization

In the initial mechanism developed above, the rate constants of the reactions in the C_4–C_5 sub-mechanism are directly taken from the detailed mechanism developed by Heufer et al. [18] and Togbe et al. [12]. Due to the removal of the isomers in the C_4–C_5 sub-mechanism, it is necessary to optimize the rate constants of the reactions in the C_4–C_5 sub-mechanism in order to improve the accuracy of the skeletal mechanism.

In the previous work [37], it was found that if a mechanism was capable of reproducing the ignition delay times in shock tubes, the mechanism can also satisfactorily predict the ignition delay times in RCMs under the same operating conditions. Meanwhile, if a mechanism predicted the species concentrations well in JSRs, the concentrations of these species in flow reactors, premixed laminar flames, and counter-flow flames can also be reproduced satisfactorily by the mechanism. Moreover, the C_0–C_3 species evolutions are dominantly controlled by the C_0–C_3 sub-mechanism, and the laminar flame speeds of fuels are primarily sensitive to the reactions involving the C_0–C_2 species [29, 54]. Thereby, the optimization of the skeletal mechanism was specially focusing on the ignition delay times in shock tubes and the fuel consumption in JSRs over wide ranges of operating conditions by tuning the Arrhenius pre-exponential factors in the skeletal C_4–C_n sub-mechanism. However, the measured temperature range in shock tubes does not cover that in the RCM experiments for *n*-pentanol. Thus, to extend the application range of the final mechanism, the experimental data in RCMs are also employed for the mechanism optimization. Meanwhile, because the optimization of the skeletal C_4–C_5 sub-mechanism influences the reactivity of the fuel, which further affects the formation rate of the final products, the concentrations of the major products including CO, CO_2, and H_2O in JSRs are also considered in the optimization process.

In our previous study [37], the optimization of the reaction rates was performed by manual iteration based on the results of the sensitivity analysis and the path analysis, which is time-consuming and strongly depends on the experience of handlers. To solve the problem, a multi-objective genetic algorithm, NSGA-II, developed by Deb et al. [55] is employed to optimize the reaction rate constants automatically in this work. By introducing the crowding distance and elitism, NSGA-II can reduce the computational complexity and preserve the best solution effectively. To improve the robustness of the optimized mechanism and effectively handle the data, the real coding in NSGA-II is used.

The steps in the optimization process of the reaction rates of the fuel-related sub-mechanism in NSGA-II are summarized as follows. (1) *N* mechanisms are generated randomly within the uncertainty of the reaction rate constants as the initial chromosome population; (2) The ignition delay times in shock tubes and RCMs and the major species concentrations in JSRs of the initial mechanisms are calculated using the CHEMKIN-III code [56]; (3) The objective function and the crowding-distance of the initial mechanisms are calculated based on the results obtained from Step (2); (4) The new population including *N* updated mechanisms is generated by exchanging the genes based on the crossover operator. In this process, the mutation may happen to add the population diversity; (5) The CHEMKIN-III code [56] is called again to calculate the ignition delay times and the major species concentrations using the new mechanisms; (6) The objective function and the crowding-distance of the new population are calculated according to the predictions in Step (5); (7) The initial population and the new population are merged; (8) In the elitism strategy, N best mechanisms are selected from the merged population as the next generation; (9) Steps (2)–(8) are repeated until the predefined maximum generation is reached.

From the optimization process, it can be found that the objective function plays an important role. In the present study, the objective function is constructed by integrating the work of Elliott et al. [57] and Wang et al. [58] as follows:

$$f_{\text{obj}} = \sum_{i=1}^{N} \left| \frac{\eta_i^{\text{pre}} - \eta_i^{\text{exp}}}{\eta_i^{\text{exp}}} \right| + W \sum_{j=1}^{M} \left| \frac{\ln\left(A_j/A_{j,0}\right)}{\ln f_j} \right| \tag{1}$$

where N and M represent the number of the experimental data and the reactions in the C_4–C_5 sub-mechanism, respectively; η_i^{pre} and η_i^{exp} refer to the predicted and measured data, respectively; A_j is the Arrhenius pre-exponential factor of the jth reaction; $A_{j,0}$ is the initial Arrhenius pre-exponential factor of the jth reaction; f_j represents the uncertainty factor of the jth reaction, and it is set to 6 for reactions R1–R19 in the present study, which means that the variation range of the Arrhenius pre-exponential factor is from $A_{j,\,0}/6$ to $A_{j,\,0} \times 6$ in the optimization process. Different from the objective function used by Elliott et al. [57], the second term on the right-hand side of Eq. (1) is introduced to constrain the variation amplitude of the reaction rates and maximize the faithfulness of the optimized mechanism to the initial mechanism. A weight factor, W, is introduced to reduce the influence of the second term on the optimized mechanism. In the present study, W is set as 0.1 based on the overall optimization results.

Moreover, a constraint function shown in Eq. (2) is introduced to ensure the reasonability of the population and improve the convergence of the optimization progress.

$$g = \sum_{i=1}^{N} \left(\frac{\eta_i^{\text{pre}} - \eta_i^{\text{exp}}}{\sigma_i^{\text{exp}}} \right)^2 - 4000 \le 0 \tag{2}$$

where σ_i^{obs} is the measured uncertainty, and 4000 is an approximate value, which is estimated based on the calculated result of the initial mechanism using the first term of Eq. (2).

For the optimization of the n-pentanol mechanism in the present study, the ignition delay times in shock tubes and RCMs measured by Heufer et al. [42] at $T = 640$–1200 K, $p = 9$–30 bar, and $\varphi = 1.0$ are employed in Eq. (1). Moreover, the ignition delay times at $T = 700$–950 K predicted using a detailed mechanism [18] are also introduced for the mechanism optimization because only a very small number of measured data are available in the NTC zone. Meanwhile, the mole fractions of n-pentanol, CO, CO_2, and H_2O measured by Togbe et al. [12] in a JSR over the ranges of $T = 770$–1220 K, $p = 10$ atm, and $\varphi = 0.35$–2.0 are also included in Eq. (1). It should be noted that the facility effects considering the pressure increase behind the reflected shock in shock tubes and the pressure decrease after the end of the compression in RCMs are included to simulate the ignition delay times in this study. In NSGA-II, the genetic operators and the GA parameters are set with the population size of 52, the generation size of 1500, the crossover probability of 0.9, the mutation probability of 0.05, the crossover distribution index of 10, and the mutation distribution index of 20.

The evolutions of the objective functions for the ignition delay times and the species concentrations obtained from NSGA-II are illustrated in Fig. 4. As can be seen, the distribution of the two objective functions of the initial generation (generation = 1) is considerably sparse. The performance of the optimized mechanisms from generation = 10 to generation = 100 is improved significantly and the diversity of the population is enhanced because of the mutation. Finally, the population converges to the Pareto front. The discrepancy of the objective functions between generation = 1000 and generation = 1500 is very small, as shown in Fig. 4, which indicates that the generation size of 1500 is enough to obtain a convergent solutions in the present optimization process [59].

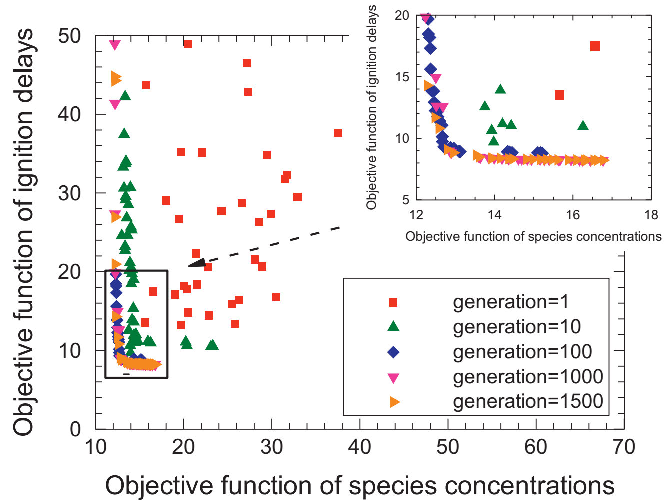

Fig. 4. Evolutions of the objective functions of the ignition delay times and the species concentrations.

A total of 52 optimized chemical mechanisms are obtained by NSGA-II. These chemical mechanisms are non-dominated by each other, which means that none of the chemical mechanism is outstanding compared to the others in the respect of minimizing the objective functions [60]. To determine the final chemical mechanism, these optimized mechanisms need to be further evaluated.

2.3. Mechanism selection

In the present research, the Method of Uncertainty Minimization using Polynomial Chaos Expansions (MUM-PCE) proposed by Sheen et al. [61] is utilized to assess the quality of the optimized mechanism. The detailed introduction of MUM-PCE is presented in Refs. [62, 63], and only a brief description is presented herein. In MUM-PCE, each rate coefficient is normalized into the factorial variable, x_i. Only the Arrhenius pre-exponential factors are optimized in this work, thus x_i can be expressed as:

$$x_i = \frac{\ln(A_i/A_{i,0})}{\ln f_i} \tag{3}$$

where A_i represents the Arrhenius pre-exponential factor of the ith reaction, $A_{i,\,0}$ and f_i represent its normal value and uncertainty factor, respectively. Here, x_i is assumed as a random variable, which is independent of each other with the uncertainty bound by $[-1, 1]$. Sheen and Wang [62] found that the uncertainty of prediction only slightly depended on the distribution of x_i. Thus, the normal distribution is used in the present study for simplicity. Regarding $\ln f_i$ as 2σ-uncertainties of $\ln A_i$, the associated polynomial chaos of x_i is expressed as:

$$x_i = \frac{1}{2}\xi_i \tag{4}$$

where ξ_i is an independent, identically distributed normal variable with the mean of 0 and the variance of 1.

In the solution mapping, the model response of each combustion property, $\eta_r(\mathbf{x})$, is expressed as [62, 64]:

$$\eta_r(\mathbf{x}) \cong \eta_{r,0} + \sum_{i=1}^{M} a_i x_i + \sum_{i=1}^{M} \sum_{j=i}^{M} b_{ij} x_i x_j \tag{5}$$

where $\eta_{r,\,0}$ represents the prediction from the initial mechanism, a_i and b_{ij} are the coefficients obtained using the Sensitivity-

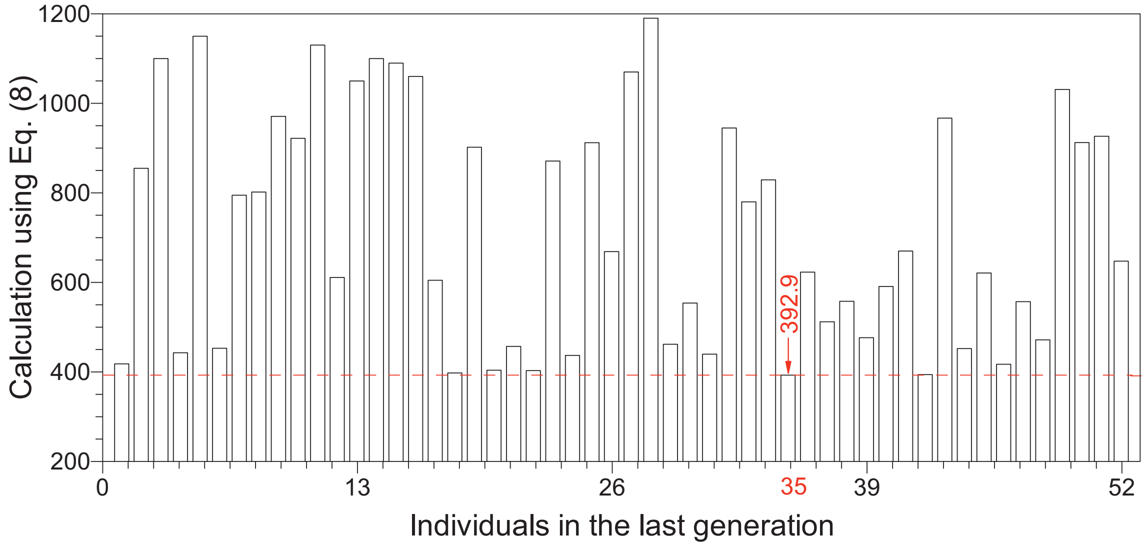

Fig. 5. Assessment on the 52 optimized mechanisms using Eq. (8).

Analysis-Based (SAB) method [65]. Substituting Eq. (4) into Eq. (5) yields:

$$\eta_r(\xi) \cong \eta_{r,0} + \sum_{i=1}^{M} \frac{1}{2} a_{r,i}\xi_i + \sum_{i=1}^{M}\sum_{j=i}^{M} \frac{1}{4} b_{r,ij}\xi_i\xi_j \tag{6}$$

Then, the uncertainty of the prediction can be obtained by taking the expectation of Eq. (6):

$$\sigma_r^2(\mathbf{x}) \cong E\left[\eta_r^2(\xi)\right] - E^2[\eta_r(\xi)] = \sum_{i=1}^{M}\left(\frac{1}{2}a_{r,i}\right)^2 + 2\sum_{i=1}^{M}\left(\frac{1}{4}b_{r,ii}\right)^2 + \sum_{i=1}^{M-1}\sum_{j=i+1}^{M}\left(\frac{1}{4}b_{r,ij}\right)^2 \tag{7}$$

In uncertainty minimization, the mechanism accuracy and precision are constrained by a set of experimental data with the least-squares optimization:

$$\Phi\left(\mathbf{x}_0^*\right) = \min_x \left\{ \sum_{r=1}^{N}\left(\frac{\left[\eta_r(\mathbf{x}) - \eta_r^{\mathrm{obs}}\right]}{\sigma_r^{\mathrm{obs}}}\right)^2 + \sum_{i=1}^{M} 4x_i^2 \right\} \tag{8}$$

where $\mathbf{x}_0^*$ refers to the optimized solution, η_r^{obs} and σ_r^{obs} represent the measured value and its uncertainty, respectively. Through the least-squares optimization, the optimized mechanism can be obtained [58, 63, 66, 67]. From Eq. (8), it can be found that the optimization process strongly depends on the generation of the response surface, i.e., the construction of $\eta_r(\mathbf{x})$. However, $\eta_r(\mathbf{x})$ might result in a large error in Eq. (8) with the highly structured surface. As found by Elliott et al. [68], the error will simultaneously affect the optimization procedure. Moreover, multiple local valleys and hills exist in Eq. (8), which produce a large amount of indistinguishable solutions [69].

In this study, $\mathbf{x}_0^*$ is determined by evaluating the optimized solutions in the Pareto front obtained from the genetic algorithm, i.e., the individuals in generation = 1500 in Section 2.3, using Eq. (8). A smaller value of Φ means a better performance of mechanism. As can be seen from Fig. 5, the value of the individual 35 is the smallest, thus it is chosen as the final optimized solution.

The uncertainty in the optimized model, $\mathbf{x}^*$, is estimated by a multivariate normal distribution:

$$\mathbf{x}^* = \mathbf{x}_0^* + \sum_{i=1}^{M} \alpha_i^* \xi_i \tag{9}$$

with mean $\mathbf{x}_0^*$ and covariance matrix $\boldsymbol{\Sigma} = \alpha^{*\mathrm{T}}\alpha^*$. By substituting the linearized response surface in the neighborhood of $\mathbf{x}_0^*$ into the joint probability density function of $\mathbf{x}$, the covariance matrix, $\boldsymbol{\Sigma}$, can be obtained:

$$\boldsymbol{\Sigma} = \alpha^{*\mathrm{T}}\alpha^* = \left[\sum_{r=1}^{N} \frac{\mathbf{W}_r\left(\mathbf{x}_0^*\right)}{\left(\sigma_r^{\mathrm{obs}}\right)^2} + 4\mathbf{I}\right]^{-1} \tag{10}$$

with

$$\mathbf{W}_r(\mathbf{x}_0^*) = \left(\mathbf{b}_r + \mathbf{b}_r^{\mathrm{T}}\right)\mathbf{x}_0^*\mathbf{x}_0^{*\mathrm{T}}\left(\mathbf{b}_r + \mathbf{b}_r^{\mathrm{T}}\right) + \mathbf{a}_r\mathbf{x}_0^{*\mathrm{T}}\left(\mathbf{b}_r + \mathbf{b}_r^{\mathrm{T}}\right) + \left(\mathbf{b}_r + \mathbf{b}_r^{\mathrm{T}}\right)\mathbf{x}_0^*\mathbf{a}_r^{\mathrm{T}} + \mathbf{a}_r\mathbf{a}_r^{\mathrm{T}} \tag{11}$$

where $\mathbf{I}$ represents the identity matrix. The parameter uncertainties, $\boldsymbol{\alpha}^*$ is minimized through the Cholesky factorization of $\boldsymbol{\Sigma}$. By substituting $\boldsymbol{\alpha}^*$ into Eqs. (5) and (7), the uncertainty of the predicted results from the final optimized mechanism can be obtained.

The predicted uncertainties of the ignition delay times using the initial and final mechanism are compared with the experimental data in Fig. 6. In the simulation, the experimental data from Togbe et al. in a JSR [12] and Heufer et al. in a shock tube and a RCM [42] are employed to constrain the predicted uncertainty of the final optimized mechanism. The uncertainty is set to 15% for all the measured results based on Refs. [12, 70]. It can be found that the uncertainties are more obvious in the low-temperature and NTC zones than that in the high-temperature zone for both the initial and optimized mechanisms. This results from the larger sensitivity coefficients of the reactions in the fuel-related sub-mechanism in the low-temperature and NTC regions than that at high temperatures (see Fig. 8). Furthermore, the predicted uncertainties using the initial mechanism are much larger than that of the final mechanism, because the uncertainties are constrained effectively by the measured data through the optimization process. Due to less experimental data available at the low-temperature conditions, the predicted uncertainty of the optimized mechanism is still relatively large at low temperatures. Thus, more experimental data at low temperatures would be beneficial for the refinement of the optimized mechanism.

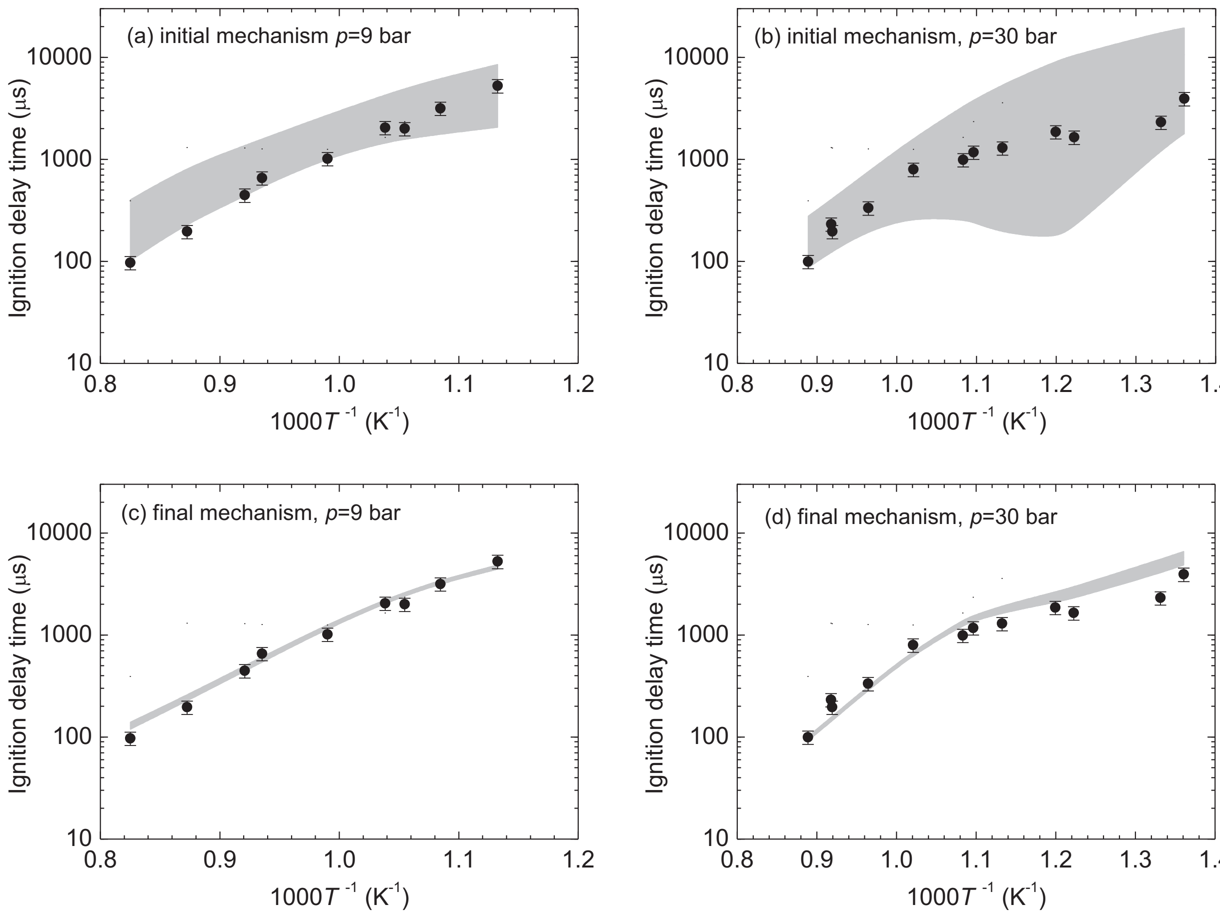

Fig. 6. Comparisons of the measured and calculated ignition delay times using the initial and final mechanism in a shock tube for $p = 9$ and 30 bar at $\varphi = 1.0$ (symbols are the experimental data from Heufer et al. [18], and light bands are the 2σ uncertainty of the model prediction).

3. Mechanism validation

3.1. Ignition delay time in shock tube and RCM

Heufer et al. [42] measured the ignition delay times of the stoichiometric n-pentanol/air mixture in a high-pressure shock tube and a RCM at $p = 9$–30 bar and $T = 640$–1200 K. The experiment was simulated using a homogeneous adiabatic model. In the simulation, the non-ideal facility effects were included by introducing a volume profile deduced from the measured non-reacting pressure profiles. The ignition delay time is defined as the time at which the maximal pressure gradient is reached.

Figure 7 displays comparisons of the experimental data and the predicted results using the initial mechanism and the final mechanism for ignition delay times. As shown in Fig. 7, the initial mechanism overpredicts the ignition delay times obviously over the whole operating condition, especially at low temperatures and in the NTC zone because of the elimination of the isomers of the C_4–C_5 species. Thus, it is necessary to tune the rate constants of the related reactions to reproduce the ignition delay times over low-to-high temperatures. Through optimizing the Arrhenius pre-exponential factors of reactions R1–R19 using the genetic algorithm, the accuracy of the mechanism is improved significantly, as can be seen from the good agreement between the measured data and the calculated results using the final mechanism as shown in Fig. 7. However, there are still some discrepancies between the predictions and the measurements on the ignition delay times in the NTC region at 30 bar. Because the predictions of the ignition behaviors from the improved mechanism and the detailed mechanism [18] are very close at 30 bar, more experimental data at high pressure would be helpful for further improvement of the n-pentanol mechanism.

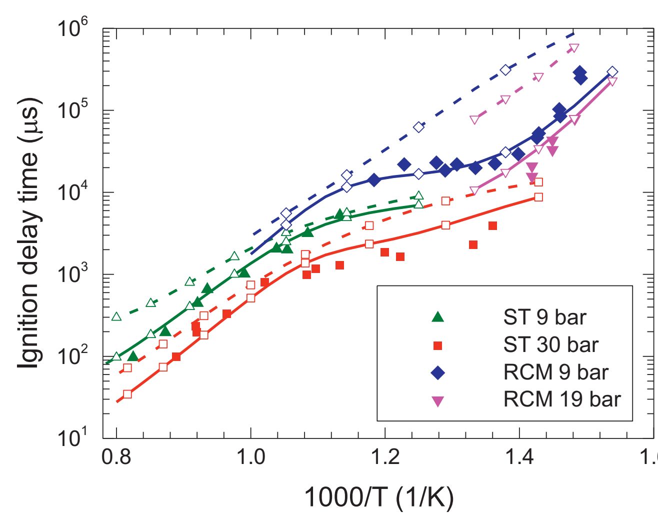

Fig. 7. Comparisons of the measured and calculated ignition delay times using the initial and optimized mechanism at $\varphi = 1.0$ for n-pentanol/air mixture (symbols are the experimental data from Heufer et al. [18], solid lines are the predictions of the final mechanism, and dashed lines are the predictions of the initial mechanism).

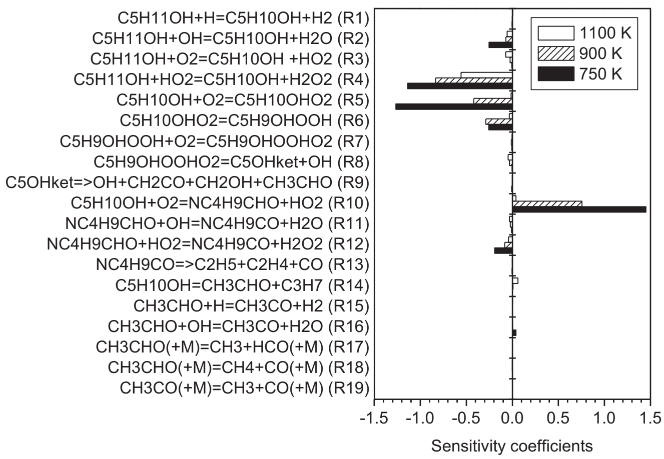

Fig. 8. Sensitivity coefficients of reaction R1–R19 for the ignition delay times in shock tube at $p = 30$ bar and $\varphi = 1.0$.

A sensitivity analysis of the optimized mechanism is performed on the ignition delay time of *n*-pentanol/air mixture in a shock tube at $T = 750$, 900, and 1100 K, $p = 30$ bar, and $\varphi = 1.0$. The sensitivity coefficients of each reaction are calculated using Eq. (12) [71]:

$$\sigma = \frac{\log \tau_+ - \log \tau_-}{\log 2 - \log 0.5} \tag{12}$$

where σ represents the sensitivity coefficient, τ_+ and τ_- are the ignition delay time calculated using the initial mechanism with the reaction rate multiplied by a factor of 2 and 0.5, respectively. Because the rate constants of reactions R1–R19 are optimized in this study and these reactions play a dominant role on the ignition characteristics of *n*-pentanol, only the sensitivity coefficients of reactions R1–R19 are shown in Fig. 8. It can be found that the sensitivity coefficients of the reactions are larger in the low-temperature ($T = 750$ K) and NTC ($T = 900$ K) regions than that in the high-temperature ($T = 1100$ K) region. Thus, by optimizing the rate constants of the reactions with the large sensitivity coefficient, the performance of the optimized mechanism is improved considerably, especially in the low-temperature and NTC regions (see Fig. 7). Moreover, the reactions of $C_5H_{10}OH + O_2 = C_5H_{10}OHO_2$ (R5) and $C_5H_{10}OH + O_2 = nC_4H_9CHO + HO_2$ (R10) demonstrate very high sensitivity coefficients with the opposite sign, indicating the competition between reaction R5 and R10. The competition reduces the reactivity of *n*-pentanol in the low-temperature and NTC zones, which results in the lower reactivity of *n*-pentanol than that of *n*-pentane [42].

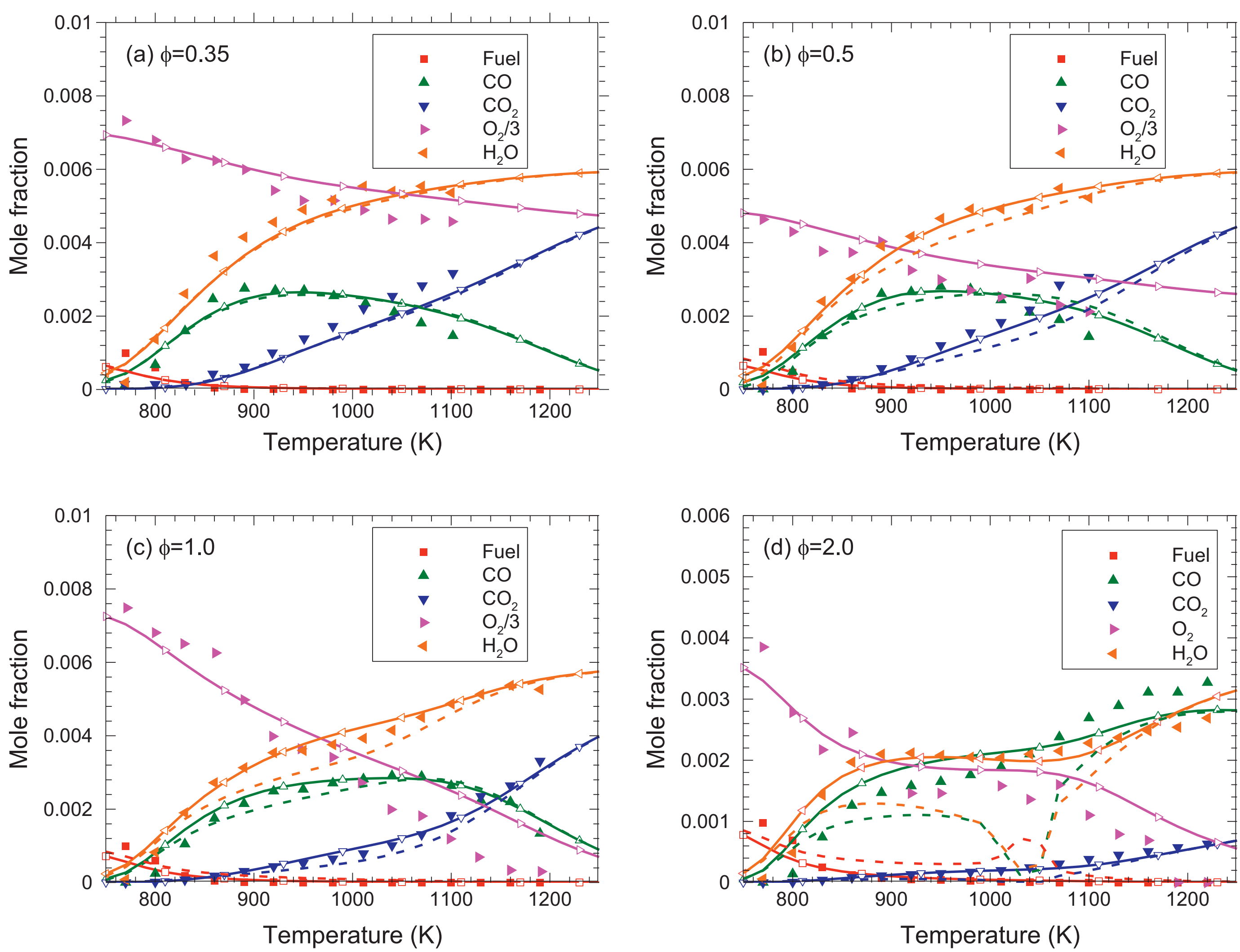

Fig. 9. Comparisons of the measured and calculated major species concentrations using the initial and optimized mechanisms in a JSR at $p = 10$ bar and $\tau = 0.7$ s with 1000 ppm *n*-pentanol/O_2/N_2 mixture for (a) $\varphi = 0.35$, (b) $\varphi = 0.5$, (c) $\varphi = 1.0$, and (d) $\varphi = 2.0$ (symbols are the experimental data from Togbe et al. [12], solid lines and symbols are the predictions of the final mechanism, and dashed lines are the predictions of the initial mechanism).

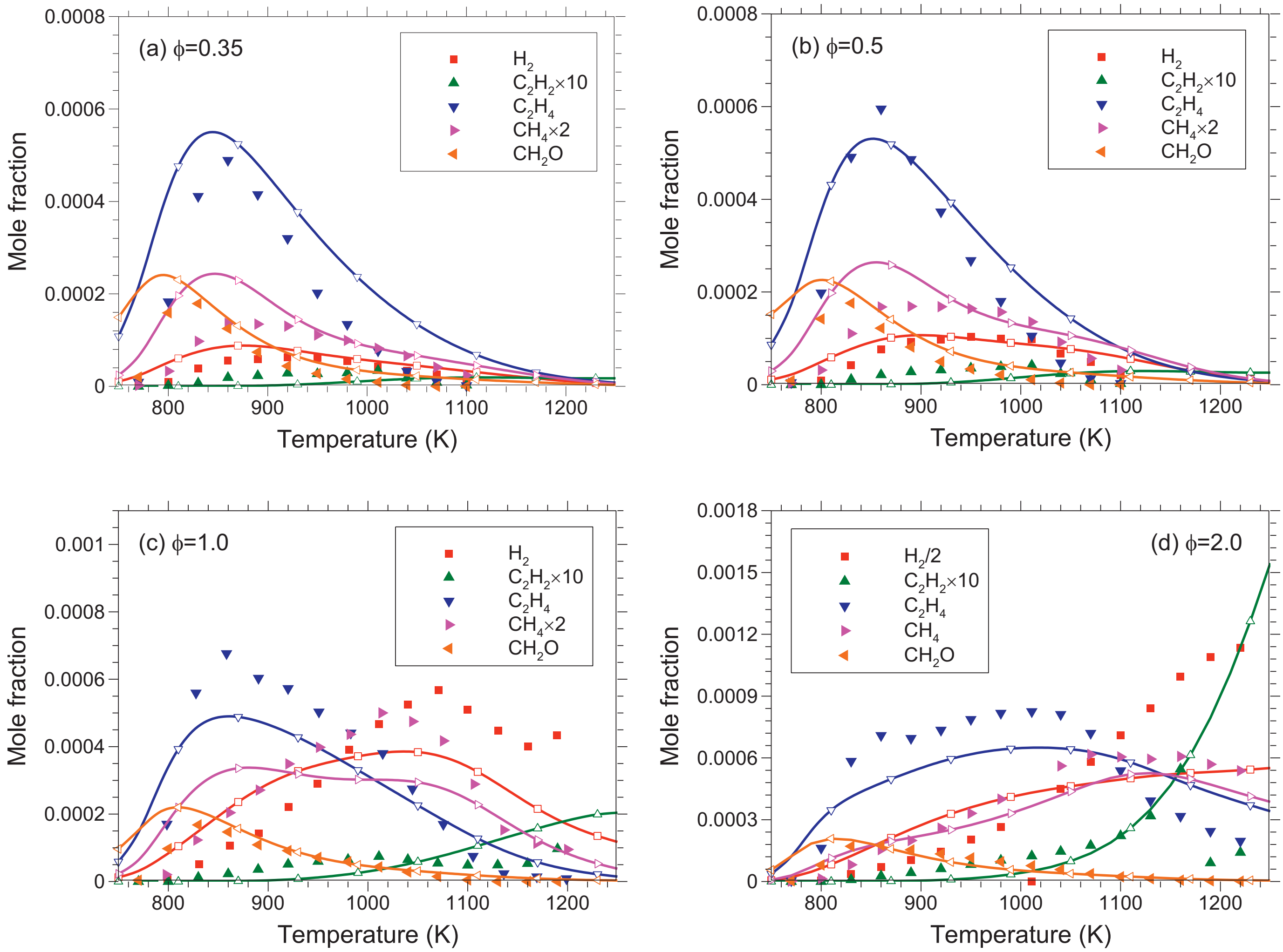

Fig. 10. Comparisons of the measured and calculated major C_0–C_2 species concentrations using the optimized mechanism in a JSR at $p = 10$ bar and $\tau = 0.7$ s with 1000 ppm *n*-pentanol/O_2/N_2 mixture for (a) $\varphi = 0.35$, (b) $\varphi = 0.5$, (c) $\varphi = 1.0$, and (d) $\varphi = 2.0$ (symbols are the experimental data from Togbe et al. [60], and solid lines and symbols are the predictions of the final mechanism).

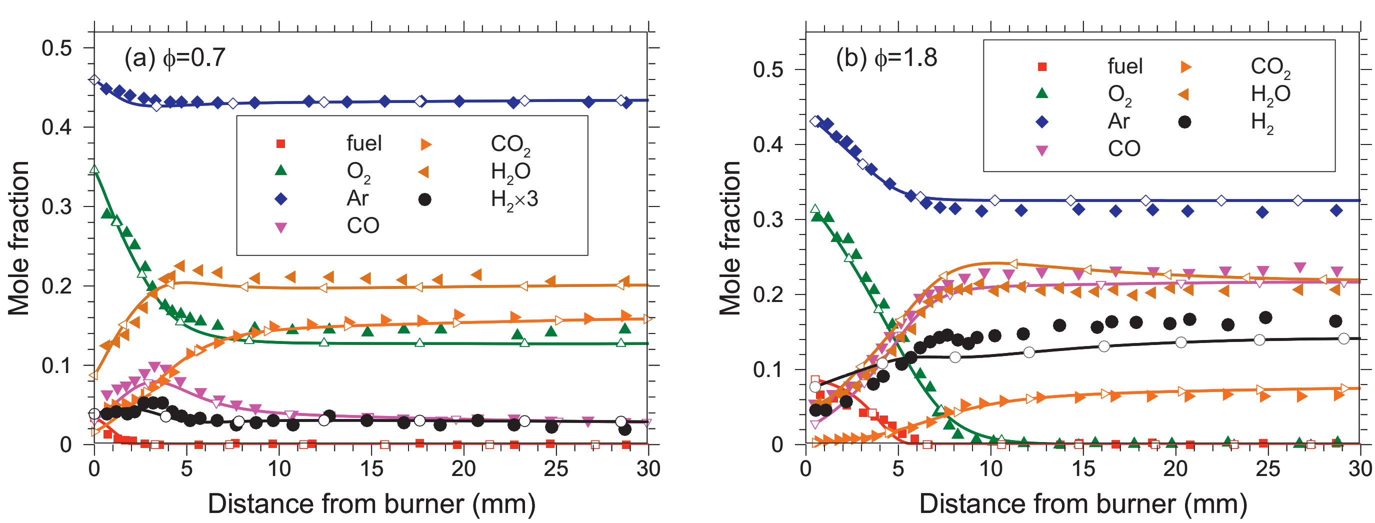

Fig. 11. Comparisons of the measured and calculated major species concentrations using the optimized mechanism in a premixed laminar flame at $p = 30$ Torr for (a) $\varphi = 0.7$ and (b) $\varphi = 1.8$ (symbols are the experimental data from Wang et al. [19], and lines and symbols are the predictions of the final mechanism).

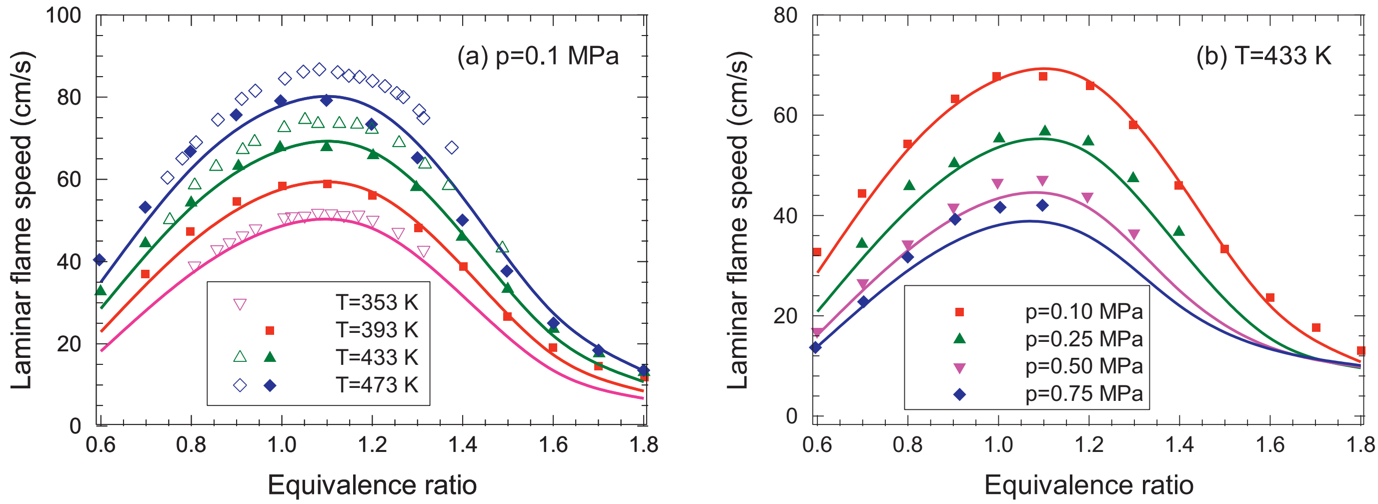

Fig. 12. Comparisons of the measured and calculated laminar flame speeds of *n*-pentanol/air using the optimized mechanism at (a) $p = 0.1$ MPa atm and (b) $T = 433$ K (solid symbols are the experimental data from Li et al. [14,78], hollow symbols are the experimental data from Nativel et al. [16], and lines are the predictions of the optimized mechanism).

Table 1
Engine specifications and operating conditions.

Parameter	Value
Engine type	Single-cylinder
Bore × Stroke	105 mm × 125 mm
Bowl volume	6.5 L
Compression ratio	16:1
IVC[a]	−133· ATDC[d]
EVO[b]	125· ATDC[d]
Engine speed	1500 rev/min
Intake air pressure (bar)	1.55
Intake air temperature (°C)	100
Equivalence ratio	0.3
EGR[c]	0–40%

[a] IVC: intake valve closing.
[b] EVO: exhaust valve opening.
[c] EGR: exhaust gas recirculation.
[d] ATDC: after top dead center.

3.2. Species concentrations in JSR

The oxidation of *n*-pentanol was measured in a JSR by Togbe et al. [12] at $T = 770–1220$ K, $p = 10$ atm, $\varphi = 0.35–2.0$, and residence time (τ) of 0.7 s with initial fuel concentration of 1000 ppm. The experiment was simulated using the perfectly stirred reactor (PSR) module in CHEMKIN-PRO [72]. Comparisons between the experimental data and the predicted results using the initial mechanism and the final mechanism are shown in Figs. 9 and 10.

Different from the results in Fig. 7, the improvement of the final mechanism on predicting the concentrations of the fuel and the major products in JSR is insignificant (see Fig. 9), except for the condition of $\varphi = 2.0$ shown in Fig. 9(d). It should be noted that no low-temperature and NTC behaviors are observed in JSR under the investigated operating conditions (see Fig. 9). At high temperatures, the fuel is consumed quickly to form small species [12, 43, 73]. Thus, the reactivity of the fuel is dominated by the small-molecule sub-mechanism at high temperatures. Due to the employment of the detailed $H_2/CO/C_1$ sub-mechanism, the agreement between the measurements and the predicted results using the initial mechanism on the major products is satisfactory. The slight discrepancy between the predictions using the initial mechanism and the final mechanism is primarily caused by the calibration of the reaction rate constants of the H-abstraction reactions (R1–R4) and the decomposition reaction (R14) in the final mechanism, which affects the radical pool and the reactivity of the fuel considerably [16, 54, 74]. Especially, at $\varphi = 2.0$ as shown in Fig. 9(d), the initial mechanism underestimates the reactivity of *n*-pentanol at $T < 1100$ K, while the predictions of the final mechanism are improved significantly.

Furthermore, the concentrations of several important $C_0–C_2$ intermediate species predicted by the final mechanism are compared with the measured data in Fig. 10. It can be found that the final mechanism reproduces the evolutions of the $C_0–C_2$ species reasonably well, although the concentrations of these species are not considered as the optimization objective in the genetic algorithm. This demonstrates the reliability of the detailed $H_2/CO/C_1$ and reduced $C_2–C_3$ sub-mechanisms used in this study. However, some discrepancies between the predictions and the measurements still exist in the H_2 and C_2H_2 concentrations at $\varphi = 1.0$ and $\varphi = 2.0$. The underestimation of the H_2 concentration at the temperature above 1000 K in the final mechanism is similar to that of the original detailed $H_2/CO/C_1$ mechanism [12, 18]. For C_2H_2, the lack of the polycyclic aromatic hydrocarbon (PAH) sub-mechanism might be responsible for the overprediction of its concentration at high temperatures and high equivalence ratios [46, 75, 76].

3.3. Species concentrations in premixed laminar flame

The premixed laminar flame of *n*-pentanol/O_2/Ar mixture was measured by Wang et al. [19] at $p = 30$ Torr, $\varphi = 0.7$ and 1.8. The experiment was simulated using the PREMIX module in CHEMKIN-PRO [72] in this section. In the simulation, the measured temperature profile was used as the input parameter, and the temperature profile was slightly shifted to remove the probe effects [77].

The measured major species concentrations are compared with the results predicted by the final mechanism in Fig. 11. The results indicate that the overall prediction error of the final mechanism is within 15% relative to the experimental data. For both equivalence ratios, the fuel consumption is reproduced satisfactorily by the final mechanism, which implies that the present skeletal mechanism well captures the major paths of the *n*-pentanol oxidation at the test conditions. Due to the employment of the detailed $H_2/CO/C_1$ sub-mechanism, the agreements between the measured and predicted O_2, CO, CO_2, and H_2O concentrations are also good. However, some discrepancies between the measurements and the predictions can be found on the H_2 concentration at $\varphi = 1.8$ in Fig. 11(b), though the H_2 concentration is predicted fairly well at $\varphi = 0.7$ (see Fig. 11(a)). The analysis based on the rate of

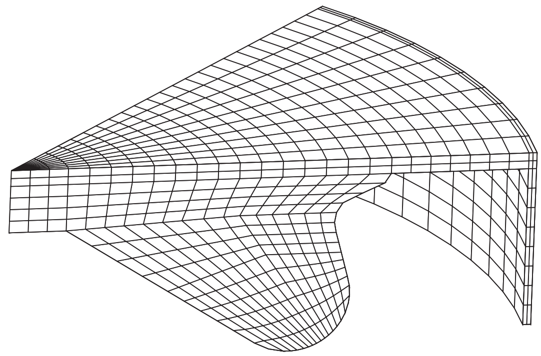

Fig. 13. Computational mesh at top dead center for engine simulation.

production indicates that H_2 is primarily formed via the H-abstraction reactions by H atom. In order to keep the compact size of the final mechanism, a number of reactions are removed in the present C_2–C_5 sub-mechanism, which might result in the above discrepancy. Further refinement of the C_2–C_3 sub-mechanism could improve the prediction of the H_2 concentration.

3.4. Laminar flame speed

Li et al. [14, 78] measured the laminar flame speed of *n*-pentanol/air mixture in a constant-volume chamber over the temperature range of 393–473 K, the pressure range of 0.10–0.75 MPa, and the equivalence ratio range of 0.6–1.8. Recently, Nativel et al. [16] determined the laminar flame speed of the *n*-pentanol/air using a heated spherical bomb at $T = 353$–473 K, $p = 0.1$ MPa, and $\varphi = 0.7$–1.5. In this study, the above experiments were simulated using the PREMIX module with the mixture-averaged transport model and the thermal diffusion effect.

Figure 12 compares the measured and predicted laminar flame speed using the final mechanism at various pressures and temperatures. As can be seen, the laminar flame speed of the *n*-pentanol/air mixture increases with the increasing equivalence ratio and reaches the peak at $\varphi = 1.1$, then it decreases monotonically when the equivalence ratio further increases. The final mechanism reproduces the trend rather well. Moreover, the increases of the laminar flame speed with the increases of the initial temperature and the decreases of the initial pressure are also well reproduced by the final mechanism. Overall, the final mechanism is capable of predicting the flame propagation characteristics rather well under wide operating conditions, and only a slight discrepancy is found for $\varphi > 1.1$ at $p = 0.1$ MPa and $T = 473$ K in Fig. 8(a). However, it should be noted that an obvious discrepancy is observed between the measurements of Li et al. [14, 78] and Nativel et al. [16]. Considering the uncertainties in the measurements, it can be concluded that the agreement between the measured and predicted results is satisfactory over the whole experimental conditions.

3.5. HCCI engine

The experiment of a Homogeneous Charge Compression Ignition (HCCI) engine fueled with *n*-pentanol was conducted by Liu et al. [79] in a modified single-cylinder diesel engine. The engine specifications and operation conditions are listed in Table 1. In the experiment, the in-cylinder pressures, heat release rates, and emissions were measured. The KIVA 3 V code [80, 81] was integrated with the CHEMKIN-III code [56] using the final mechanism to simulate the HCCI working process in this study. To enhance the prediction accuracy, several improvements on the physical and chemical sub-models were introduced to KIVA-3V. To avoid over-predicting turbulent energy in the re-normalization group (RNG) turbulence model, the new generalized re-normalization group (gRNG) κ–ε turbulence model proposed by Wang et al. [82] was employed. An updated heat transfer model [83, 84] was employed to describe the heat transfer process from the cylinder wall. The complex chemical kinetic combustion process was described by coupling the CHEMKIN III code [56] with the KIVA-3V code. In the simulation, a 45°sector mesh composed of 9560 grid cells was employed, as shown in Fig. 13.

Figures 14 and 15 compare the measured and predicted in-cylinder pressures, heat release rates, and CO and HC emissions. The predicted results show a reasonably good agreement with the experimental data. The increase of the exhaust gas recirculation (EGR) rate leading the longer ignition delay is well reproduced by the present model using the final mechanism, as shown in Fig. 14. For CO emissions, the influence of the EGR rate is insignificant at the EGR rates below 20%, and it increases rapidly at the EGR rates higher than 20%. The computational model well captures the trend, and only a slight overprediction of the CO emissions exists at EGR rate = 40%. For HC emissions, the agreement between the predicted and measured data is satisfactory, though the HC emissions are underestimated at EGR below 10%. Considering the errors of the experimental data, as well as the uncertainties of the input parameters and the sub-models in the computational model, it can be concluded that the present model with the employment of the op-

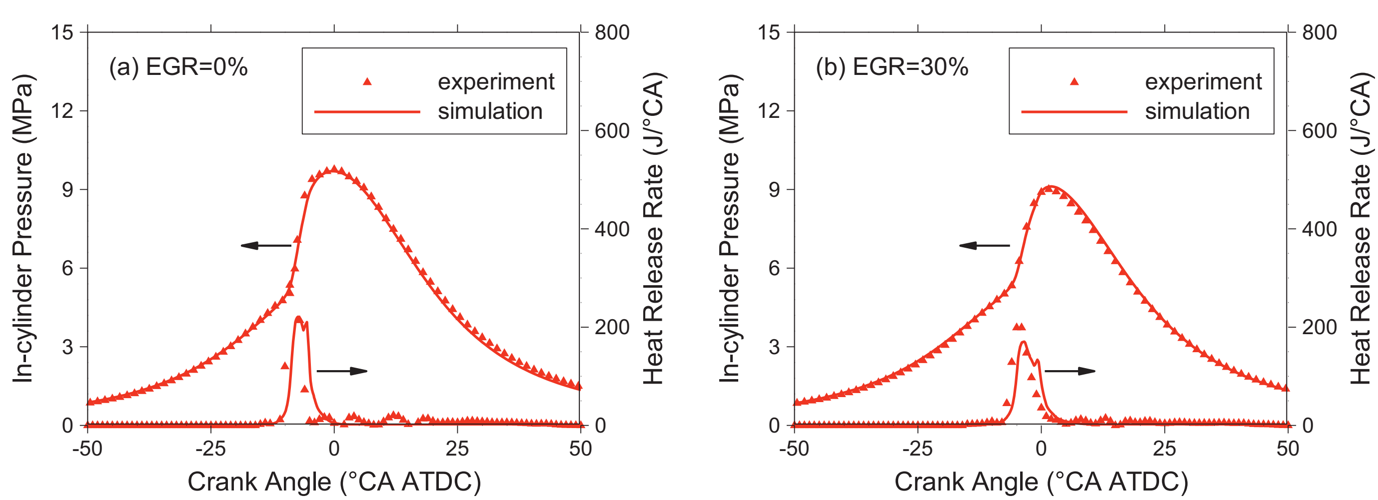

Fig. 14. Comparisons of the measured (symbols) [79] and calculated (lines) in-cylinder pressure and heat release rate in a HCCI engine fueled with *n*-pentanol under (a) EGR = 0% and (b) EGR = 30%.

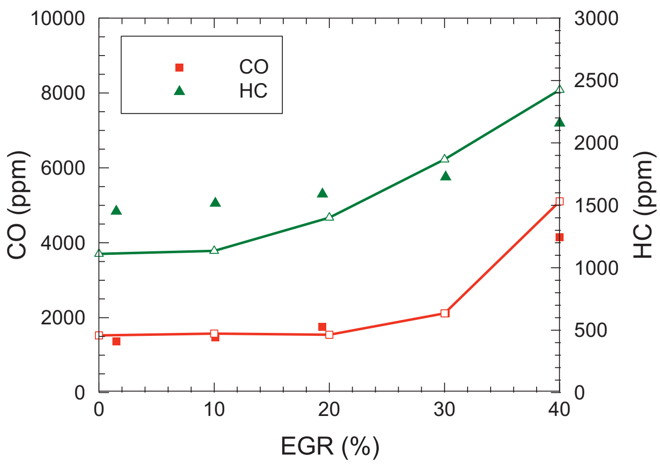

Fig. 15. Comparisons of the measured (symbols) [79] and calculated (lines) CO and HC emissions in a HCCI engine fueled with *n*-pentanol under various EGR rates.

timized *n*-pentanol mechanism is capable of reproducing the combustion and emission characteristics of HCCI engine.

4. Conclusions

A skeletal *n*-pentanol oxidation mechanism is developed by integrating the decoupling methodology, genetic algorithm, and uncertainty quantification. First, an initial skeletal mechanism including a skeletal C_4–C_5 sub-mechanism, a reduced C_2–C_3 sub-mechanism, and a detailed H_2/CO/C_1 sub-mechanism is developed based on the decoupling methodology. Then, the Arrhenius pre-exponential factors of the reactions in the C_4–C_5 sub-mechanism are automatically optimized using the non-dominated sorting genetic algorithm II (NSGA-II) to reproduce the ignition delay times in shock tubes and RCMs and the major species concentrations in JSRs over low-to-high temperatures, and a set of optimized mechanisms are obtained. The final mechanism is determined by assessing the optimized mechanisms using the method of uncertainty minimization using polynomial chaos expansions (MUM-PCE).

The final mechanism includes 47 species and 177 reactions. The comparisons between the measured and calculated results demonstrate that the final mechanism is capable of satisfactorily predicting the ignition delay times in shock tubes and RCMs, the species evolutions in JSRs and premixed laminar flames, the laminar flame speeds, and the combustion and emissions of HCCI engine under wide operating conditions. Therefore, the integrating method proposed in this study is reliable and effective, and it can be extended to the construction of skeletal mechanisms of other long-chain hydrocarbon fuels.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2018.04.012.

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